

Geomechanical Borehole Stability & Pore Pressure Prediction in Upper Assam Formations

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ABSTRACT

Drilling exploratory and development wells in the Upper Assam Shelf Basin (Duliajan, Moran, and Naharkatiya fields) presents significant geomechanical challenges. The Barail Group (~1,800m–2,500m) consists of highly reactive smectite-rich shales prone to swelling and pipe entrapment. The underlying Kopili Formation (~2,500m–3,200m) exhibits sub-hydrostatic to overpressured regimes requiring equivalent mud weights up to 1.38 g/cc. Finally, the Sylhet Limestone (~3,000m–3,500m) contains extensive karst vugs leading to total lost circulation. This paper integrates wireline log signatures (GR, RHOB, DTC) and historical offset well data across 45 wells to construct a predictive geomechanical model.

KEY FORMATION REGIMES & RECOMMENDED DRILLING WINDOWS

Formation Name	Depth Interval	Pore Pressure (EMW)	Fracture Gradient	Primary Drilling Hazard
Tipam Sandstone	0 - 800 m	1.02 - 1.08 g/cc	1.65 g/cc	Washout, high permeability
Girujan Clay	800 - 1,500 m	1.10 - 1.15 g/cc	1.72 g/cc	Clay swelling, tight hole
Barail Group	1,800 - 2,500 m	1.18 - 1.25 g/cc	1.80 g/cc	■ Reactive Shale / Stuck Pipe
Kopili Formation	2,500 - 3,200 m	1.28 - 1.38 g/cc	1.88 g/cc	■ Overpressure Kick / Influx
Sylhet Limestone	3,000 - 3,500 m	1.15 - 1.22 g/cc	1.52 g/cc	■ Fractured Lost Circulation